

Unit 2

Application of Law: Cybercrime and Information Technology Act

Purpose of IT Act, 2000 and ITAA 2008

- The Information Technology Act, 2000 was enacted to give legal recognition to electronic records and transactions.
- It equates digital signatures with handwritten ones and supports e-governance and e-commerce.
- The 2008 Amendment (ITAA 2008) introduced stronger data protection, cyber terrorism, and identity theft provisions.
- It also expanded the jurisdiction of the Act to cover offences committed abroad that affect systems in India.
- Together, these Acts form the backbone of India's cyber law framework.

Scope and Jurisdiction

- Under Section 75 of ITAA 2008, offences committed outside India are punishable if a computer system in India is involved.
- This ensures Indian jurisdiction over global cybercrimes targeting Indian systems or data.
- Certain paper-based transactions like wills and property deeds are excluded from ITA's scope.
- Electronic record retention and tracking are legally mandated.
- These provisions enable investigation and prosecution even across borders.

Relevant Legal Provisions

Key sections include:

- Sec 43 & 66: Unauthorized access, damage, and hacking.
- Sec 66C–66E: Identity theft, cheating by impersonation, and privacy violation.

- Sec 67–67B: Obscenity and child pornography.
 - Sec 69–70B: Interception, monitoring, and CERT-In's powers.
- They define offences, empower investigation, and specify penalties.

Cyber Crimes Mapping with ITAA 2008, IPC and Special & Local Laws

Sl. No.	Nature of Complaint	Applicable Section(s) and Punishments under ITAA 2008	Applicable Section(s) under Other Laws (IPC/Other Acts)
1	Mobile phone lost / stolen	—	Section 379 IPC – Up to 3 years imprisonment or fine or both
2	Receiving stolen computer/mobile phone/data	Section 66B ITAA 2008 – Up to 3 years imprisonment or ₹1 lakh fine or both	Section 411 IPC – Up to 3 years imprisonment or fine or both
3	Data owned by you or your company is stolen	Section 66 ITAA 2008 – Up to 3 years imprisonment or ₹5 lakh fine or both	Section 379 IPC – Up to 3 years imprisonment or fine or both
4	A password is stolen and used by someone else for fraudulent purpose	Section 66C & 66D ITAA 2008 – Up to 3 years imprisonment and ₹1 lakh fine	Section 419 IPC – Up to 3 years imprisonment; Section 420 IPC – Up to 7 years imprisonment and fine
5	An e-mail is read by someone else using your password	Section 66 & 66C ITAA 2008 – Up to 3 years imprisonment or ₹5 lakh fine	—
6	A biometric thumb impression is misused	Section 66C ITAA 2008 – Up to 3 years imprisonment or ₹1 lakh fine	—
7	An electronic or digital signature is misused	Section 66C ITAA 2008 – Up to 3 years imprisonment or	—

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		₹1 lakh fine	
8	Phishing e-mail sent in your name asking for login credentials	Section 66D ITAA 2008 – Up to 3 years imprisonment or ₹1 lakh fine	Section 419 IPC – Up to 3 years imprisonment or fine or both
9	Capturing/publishing / transmitting private images without consent	Section 66E ITAA 2008 – Up to 3 years imprisonment or ₹2 lakh fine	Section 292 IPC – Up to 2 years imprisonment and fine ₹2000 (5 years and ₹5000 for repeat offence)
10	Tampering with computer source documents	Section 65 ITAA 2008 – Up to 3 years imprisonment or ₹2 lakh fine	—
11	Data modification	Section 66 ITAA 2008 – Up to 3 years imprisonment or ₹5 lakh fine	—
12	Sending offensive messages through communication service	Section 66A ITAA 2008 – Up to 3 years imprisonment and fine	Sections 500, 504, 505, 507, 508, 509 IPC – 2 to 7 years imprisonment or fine
13	Publishing or transmitting obscene material electronically	Section 67 ITAA 2008 – 3 years + ₹5 lakh (first offence), 5 years + ₹10 lakh (subsequent)	Section 292 IPC – Up to 2 years and ₹2000 fine; repeat 5 years + ₹5000
14	Publishing sexually explicit material	Section 67A ITAA 2008 – 5 years + ₹10 lakh (first), 7 years + ₹10 lakh (repeat)	Section 292 IPC – Up to 2 years + ₹2000 fine; repeat 5 years + ₹5000
15	Publishing child porn or explicit material	Section 67B ITAA 2008 – 5 years + ₹10 lakh (first), 7 years + ₹10 lakh (repeat)	Section 292 IPC – Same as above
16	Misusing Wi-Fi	Section 66F ITAA 2008 –	—

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	connection (against the State)	Life imprisonment	
17	Planting a computer virus	Section 66 ITAA 2008 – Up to 3 years imprisonment or ₹5 lakh fine	—
18	Denial of Service (DoS) attack on government system	Section 66 ITAA 2008 – Up to 3 years imprisonment or ₹5 lakh fine	—
19	Stealing data from a government system (national security)	Section 66F ITAA 2008 – Life imprisonment	—
20	Refusing to decrypt communication when ordered	Section 69 ITAA 2008 – Up to 7 years imprisonment	—
21	Intermediaries not providing access to information	Section 69 ITAA 2008 – Up to 7 years imprisonment	—
22	Failure to block websites when ordered	Section 69A ITAA 2008 – Up to 7 years imprisonment	—
23	Sending threatening messages by e-mail	Section 66A ITAA 2008 – Up to 3 years imprisonment and fine	Section 503 IPC – Up to 2 years imprisonment or fine
24	Sending messages insulting a woman's modesty	Section 66A ITAA 2008 – Up to 3 years imprisonment and fine	Section 509 IPC – Up to 1 year imprisonment or fine
25	Creating bogus websites or cyber frauds	Section 66D ITAA 2008 – Up to 3 years imprisonment or ₹1 lakh fine	Section 419 IPC – 3 years; Section 420 IPC – 7 years imprisonment
26	E-mail spoofing	Section 66C ITAA 2008 – Up to 3 years imprisonment or	Section 465 IPC – 2 years; Section 468 IPC –

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		₹1 lakh fine	7 years imprisonment
27	Making a false electronic document	Section 66D ITAA 2008 – Up to 3 years imprisonment or ₹1 lakh fine	Section 465 IPC – 2 years or fine or both
28	Forgery for purpose of cheating	Section 66D ITAA 2008 – Up to 3 years imprisonment or ₹1 lakh fine	Section 468 IPC – Up to 7 years imprisonment and fine
29	Forgery for harming reputation	Section 66D ITAA 2008 – Up to 3 years imprisonment or ₹1 lakh fine	Section 469 IPC – Up to 3 years imprisonment and fine
30	E-mail abuse	Section 66A ITAA 2008 – Up to 3 years imprisonment and fine	Section 500 IPC – Up to 2 years or fine or both
31	Criminal intimidation using e-mail	Section 66A ITAA 2008 – Up to 3 years imprisonment and fine	Section 506 IPC – Up to 7 years imprisonment and fine
32	Anonymous threats via communication	Section 66A ITAA 2008 – Up to 3 years imprisonment and fine	Section 507 IPC – Up to 2 years imprisonment
33	Copyright infringement	Section 66 ITAA 2008 – Up to 3 years imprisonment or ₹5 lakh fine	Section 63, 63B Copyright Act 1957 – Up to 3 years imprisonment or fine
34	Theft of computer hardware	—	Section 379 IPC – Up to 3 years imprisonment or fine
35	Online sale of drugs	—	NDPS Act – Imprisonment and fine as per offence
36	Online sale of arms	—	Arms Act – Punishment

			as per severity of offence
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Empowered Officers and Blocking Rules

- Police Inspectors and above can investigate offences under ITAA 2008.
- Section 69A authorizes blocking of public access to harmful online content by a Joint Secretary-level officer.
- Interception or decryption can be ordered by Home Secretaries under the 2009 Rules.
- These administrative powers help contain online threats promptly.
- CERT-In acts as the national nodal agency for cyber security incidents.

International Investigations and Letter Rogatory

Need for International Procedures

Cybercrimes often transcend borders—the offender, victim, and server may be in different countries. Indian law ends at territorial limits, making international cooperation vital. Hence, investigators use Mutual Legal Assistance Treaties (MLATs) or Letters Rogatory (LRs). These allow lawful data exchange and evidence collection from foreign authorities. Without them, international data requests are invalid in court.

Legal Provisions (CrPC Sections 166-A and 166-B)

These sections authorize Indian courts to request or respond to foreign investigation assistance.

- Under 166-A, courts can issue a Letter of Request to foreign courts for examination, evidence, or document collection.
- Under 166-B, India can honor similar requests from foreign courts.

Statements and documents obtained this way are admissible in Indian courts. The Ministry of Home Affairs coordinates such requests.

Meaning and Function of Letter Rogatory

- A Letter Rogatory is a formal judicial communication between courts of different countries.
- It seeks testimony, document production, or evidence from a foreign jurisdiction.
- It is based on the principle of mutual legal respect and comity among nations.
- The foreign court executes the request and sends back authenticated results.
- This ensures legality and reliability of international evidence.

Procedure for Sending Letter Rogatory

- Section **166-A** of the **Criminal Procedure Code, 1973** provides the legal framework for Indian investigating agencies to collect evidence or documents from foreign countries through a **Letter Rogatory** (also known as a *Letter of Request*).
- A Letter Rogatory is sent via a competent court under the framework of **Mutual Legal Assistance Treaties (MLATs)**, **Memorandums of Understanding (MoUs)**, or on a **reciprocal basis** with the concerned nation. It allows formal cooperation between India and other countries in criminal investigations.
- Before submission, **investigating agencies must obtain prior approval from the Central Authority**, i.e., the **Ministry of Home Affairs (MHA)**, Government of India. The request must be formally drafted and include all relevant evidence, photographs, and documents properly referenced and clearly labeled. All enclosures must be **legible, page-numbered, and translated** into the language required by the treaty or the requested country. The translated copies must be **authenticated and signed by a certified translator**.
- At least **five copies** of the Letter Rogatory, including the original, are to be prepared. Three copies (with translations) are sent to the **MHA**, and one copy to the **International Police Cooperation Cell of the CBI** for processing.
- Before submission, the **Investigating Officer (IO)** must obtain a **No Objection Certificate (NOC)** from the **Director/Department of Public Prosecution**, confirming that the case satisfies the **Dual Criminality Principle** (i.e., the act is a crime in both countries).

- Finally, the **Letter Rogatory and NOC** are routed through the **Interpol Liaison Officer, CBI**, ensuring proper coordination and adherence to international protocols.

Pre-Investigation Assessment

Purpose of Pre-Investigation Assessment

The pre-investigation phase is the most crucial stage in any digital forensic case. It focuses on understanding what happened, where it happened, and whether an investigation is justified. Investigators assess the situation to ensure that their actions are lawful, ethical, and necessary. This step prevents wastage of time, resources, and reputation due to premature or unnecessary investigations. Pre-assessment ensures that investigations begin with legitimate grounds and a clear understanding of context.

Before launching a forensic investigation, it is vital to verify the authenticity of the complaint or alert. False accusations, automated alarms, or internal disputes can sometimes trigger unnecessary probes. Investigators must confirm the origin, intent, and reliability of the initial report. They also evaluate whether the event truly violates policy or law.

The organization's readiness determines the effectiveness of the investigation. This includes having clear policies, trained staff, forensic infrastructure, and a communication protocol. If readiness is lacking, the organization should first stabilize systems before collecting evidence. Investigators ensure they have authorization, scope clarity, and technical access.

Legal and Ethical Considerations

- Every investigation must begin within the framework of applicable laws and ethical codes.
- Unauthorized collection of evidence or privacy intrusion can invalidate the case.
- Investigators confirm that they have legal authority or consent to proceed.
- They must act impartially, respecting both victim and suspect rights.
- Acting within legal and ethical boundaries safeguards both evidence validity and investigator reputation.

Establishing Investigation Objectives

The objectives define what needs to be proved or disproved.

- Investigators must decide whether the goal is data recovery, fraud detection, or policy violation verification.
- These objectives guide the selection of tools, methods, and team members.
- Clear objectives also prevent scope creep during later stages.
- Defined objectives ensure focus, efficiency, and clarity throughout the investigation.

Is it a crime (as per ITAA 2008) in the first place?

Understanding the Indian IT Act Framework

- The Information Technology Act, 2000 (amended 2008) governs cybercrimes and digital evidence admissibility in India.
- Before proceeding, investigators must determine whether the incident qualifies as an offence under the Act.
- The Act provides the legal foundation for investigating unauthorized access, data theft, and online frauds.
- The classification determines whether the case is civil, disciplinary, or criminal.
- The IT Act serves as the baseline for determining the legal nature and severity of a digital incident.
- Common Cyber Offences Under the IT Act
 - **Section 43** – Unauthorized access, data theft, or damage to computer systems.
 - **Section 66** – Computer-related offences involving fraud or dishonesty.
 - **Section 66C** – Identity theft or misuse of digital signatures.
 - **Section 66D** – Cheating by personation using computer resources.
 - **Section 67 & 67A** – Publishing obscene or sexually explicit content online.Each offence carries specific penalties and investigation procedures. Identifying the correct section of ITAA helps in applying the right legal framework and response.

Preliminary Review of Scene of Offence

The scene review identifies the scope of digital evidence and environmental factors. Investigators assess computers, networks, or storage devices possibly involved. They determine potential data sources and evidence volatility. The review sets the groundwork for proper evidence handling and containment. Scene review helps understand the digital ecosystem before evidence is disturbed. Typically, the scene of offence can be broadly dealt under,

1. Home of individuals with one or more computers.
2. Cyber Café/Public places.
3. Companies / organizations, with one or more computers and in some cases with vast and complicated network of systems

Initial Scene Assessment

Upon arrival, the Investigating Officer (IO) must survey and secure the scene, ensuring no one tampers with evidence. The IO should carefully observe, assess, and document the scene before proceeding to collect or seize evidence. Since digital evidence is fragile and volatile, it can exist across multiple devices such as computers, routers, fax machines, or even copiers.

Pre-Investigation Review

A preliminary review helps the IO understand the local environment, network setup, and technical context. The IO must record these observations in the questionnaire for the Forensic Science Laboratory (FSL), which assists experts in analysis. Videography, photography, and network sketches should be created to capture the scene “as-is” for reference in official records or the panchnama (scene documentation).

Securing and Documenting the Scene

The IO must secure the area and note down all persons present, along with their roles. Identify all potential evidence, both physical (manuals, notes, passwords) and digital

(devices, peripherals). Pay attention to equipment positioning (e.g., mouse placement may suggest whether the user is left- or right-handed).

Handling Fragile Digital Evidence

Identify and protect volatile data (RAM, active sessions, cache) that can easily be lost. Avoid disturbing electronic evidence — do not turn ON a powered-OFF system, and do not turn OFF a powered-ON system. Photograph and document all running systems before seizure, ensuring technical personnel are available for data recovery.

Recording Connections and Network Layout

The IO should document all network cables, power lines, modems, and telephone connections. Each connection point must be labeled at both ends (device and wall source) for accurate reconstruction later. This mapping helps forensic experts recreate the network architecture and trace data flow during analysis.

Pre-Investigation Technical Assessment

a) Purpose of Pre-Investigation Technical Assessment

The pre-investigation technical assessment is the initial stage where the **Investigating Officer (IO)** gathers all relevant technical information to understand the **scope and nature of a cyber incident**. It helps in identifying the **crime environment, systems involved, and potential sources of digital evidence** before formal investigation begins.

b) Objective and Approach

The goal is to develop a **complete picture of the incident or network setup** through structured questioning. A **standard questionnaire** is used to help IOs gather consistent and comprehensive details about the digital environment. However, this list can be **expanded or customized** depending on the type of cybercrime or crime scene.

c) Scene of Offence: Cyber Café

- Identify the **number of computer systems** and how many are connected to the Internet.
- Obtain the **network topology or client-server structure** details.
- Collect **CCTV or webcam footage** and **Internet user logs** for the relevant period.
- Verify if **user management software** is used to track online sessions.
- Check policies on **formatting, hardware replacement, and use of removable storage media**.

Summary: The aim is to trace user activity and system usage in shared or public environments.

4) Scene of Offence: Home

- Identify the **type of Internet connection** (Wi-Fi or Ethernet).
- Determine **how many devices** are used and who has access.
- Note the **location of each system** and persons using them.
- Gather details on **external storage devices** and other peripherals such as **printers, modems, or scanners**.
- Collect information about **home network architecture**, if applicable.

Summary: Home scene assessment helps identify personal devices, users, and access patterns relevant to the crime.

5) Scene of Offence: Corporate Environment

- Two structured questionnaires guide the IO:
 - a) **Crimes using computers as instruments or repositories** (e.g., fraud, data theft).
 - b) **Crimes targeting computer systems** (e.g., hacking, denial-of-service attacks).
- These help in collecting key organizational data such as **network configuration, system logs, user access records, and security policies**.

Summary: Corporate assessment focuses on system vulnerabilities, internal misuse, and attack impact analysis.

6) Outcome of Pre-Investigation Assessment

After completing the assessment, the IO gains a **holistic understanding of the incident**—its systems, networks, and users.

This enables the IO to decide on:

- Issuing **preservation notices** to relevant persons or organizations.
- Determining the **technical expertise or forensic support** required.
- Planning the **acquisition and examination of evidence** in a structured, legally sound manner.
- The assessment equips the IO to make informed, technically accurate, and legally valid decisions on how to proceed with the investigation.

Issuance of Preservation Notice:

A preservation notice is issued by the **Investigating Officer (IO)** to ensure that **digital evidence is not deleted or tampered with** before it is formally seized or examined. The IO directs concerned parties (complainant, company, or service provider) to **stop access, take backups, and preserve logs** until further orders. This prevents the accused from altering or deleting incriminating data. Notices are sent to **all related individuals or organizations** instructing them **not to modify or erase data** relevant to the case. The process aligns with **Section 91 CrPC** and international practices ensuring lawful preservation of electronic evidence. The IO may **adapt the format** of the notice based on the case or local requirements using standard templates (Annexures 4-3 and 4-4).

Containment of the Incident / Offence

- **Containment** is the process of **limiting damage and preventing further harm** after a cyber incident or offence is reported. It ensures that additional attacks or data loss do not occur before the investigation officially begins.
- The Investigating Officer (IO) must identify the **scope and nature** of the incident and take steps to **isolate affected systems**. Common containment actions include

blocking access through firewalls, disconnecting compromised devices, disabling user accounts, or isolating suspicious network segments.

- In **financial frauds**, the IO should immediately contact banks to **freeze suspect or beneficiary accounts** to prevent withdrawal or transfer of stolen funds. For **online crimes**, such as fake or defamatory profiles, the IO should request **service providers** to **block or remove** the content and **preserve access logs** for evidence.
- If a **system must be restored** quickly for operational or public reasons, the IO should first obtain a **forensic image copy** before allowing restoration. Such actions must be **documented and justified**, as restoration usually occurs **after evidence seizure**.

In essence: Containment ensures that the cybercrime is **controlled, evidence is preserved**, and no further damage occurs while maintaining the integrity of the investigation.